Freight Rate Prediction

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1. Objective and validation design

Predict posted_rate in dollars for 12,000 unlabeled loads dated November-December 2025. The labeled development set contains 48,000 loads from January-October. Chronological evaluation simulates prediction into future months rather than mixing future observations into training.

Main development window: train January-August (38,477 loads); evaluate September-October (9,523). **Additional window:** train January-June (28,806); evaluate July-August (9,671). Both windows informed development and model selection; neither is an untouched final test. Hidden-set metrics remain unknown.

2. Data quality and preprocessing

The labeled file has 300 missing weights and 374 missing market_index values, with no exact duplicate rows or repeated load IDs in the initial inspection. Non-positive weights are invalid: 233 occur in the main training window. They are replaced with missing values, then imputed with a median fitted only on the relevant training period. Market-index imputation follows the same training-only rule for exploratory models.

Extreme prices were inspected and retained: unusual values alone do not prove label errors. load_id is excluded from predictors. One-hot encoding is fitted on training data and ignores unseen categories. Numeric inputs are standardized for Ridge and MLP; the MLP target is also standardized and predictions transformed back to dollars. Trees use unscaled inputs.

3. Models and main-window results

Model	MAE ($)	RMSE ($)	R-squared
Median rate per mile x distance	256.95	684.25	0.7990
Ridge - core features	141.46	637.45	0.8255
Random Forest - core features	124.96	633.20	0.8278
MLP - core features	142.78	667.03	0.8089
RF + market_index	146.83	643.07	0.8224
RF + market_index + quote_signal	119.20	635.45	0.8266
CatBoost - core features	182.19	652.90	0.8170
Random Forest - core + coordinates	117.29	632.24	0.8284

MAE was used as the primary local selection criterion to reflect typical dollar error; RMSE and R-squared were also reported to expose large-error sensitivity. Spotter did not disclose a final scoring metric in the supplied scorer. The market-feature model has slightly lower MAE than RF core, but worse RMSE; the coordinate model improves all three relative to RF core.

4. Selected model and temporal stability

Model	MAE ($)	RMSE ($)	R-squared
Ridge - core	141.84	628.40	0.8223
Random Forest - core	128.96	628.61	0.8222
Random Forest - core + coordinates	121.85	628.11	0.8224

Random Forest with coordinates improved MAE over RF core in both windows: 124.96 to 117.29 dollars in September-October and 128.96 to 121.85 dollars in July-August. RMSE improvements were small. The model uses 200 trees, maximum depth 18, minimum leaf size 10, max_features 0.8 and random_state 42.

Inputs are pickup, delivery, equipment, distance, weight, weekday, month and four supplied location coordinates. market_index and quote_signal are omitted from the selected model: neither is supplied for the fixed December scenario, and quote_signal provenance and prediction-time availability are undocumented. This does not establish that quote_signal is leakage.

5. Final training and December scenario

After model selection, the selected pipeline is refitted on all 48,000 labeled loads, with the weight median recomputed from that labeled set only. Validation predictions are mapped to template load_id values. December coordinates are recovered from a checked city-coordinate lookup built from the January-August labeled subset; the lookup has no conflicting coordinate pairs and contains Lexington and Fort Wayne. No external geocoding is used.

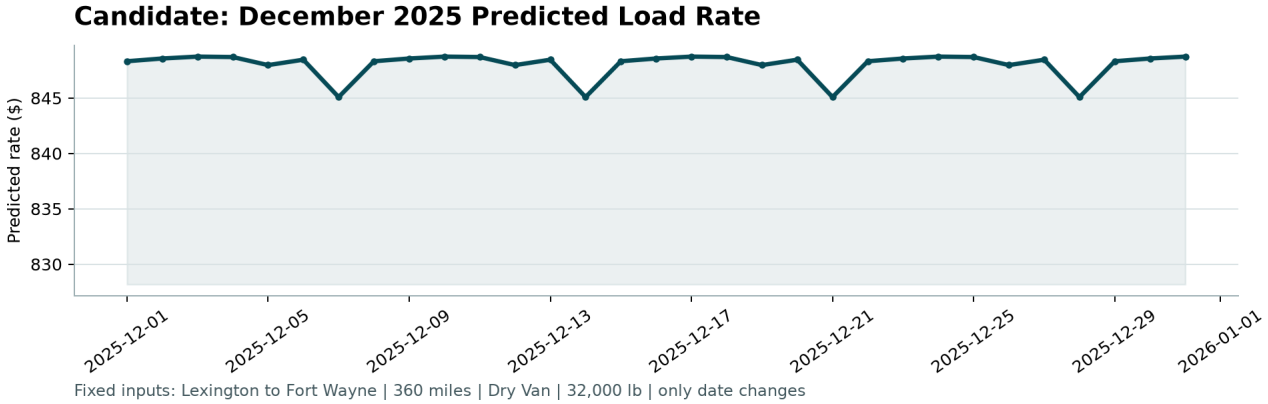


Figure: Required fixed December chart produced by the supplied score.py. Inputs remain Lexington to Fort Wayne, 360 miles, Dry Van and 32,000 lb. The weekly pattern reflects weekday effects. December is unseen during labeled training; this is not a validated holiday or December-seasonality forecast. Shading is decorative, not a confidence interval.

6. Output checks and limitations

The executed notebook reports successful scorer validation of exactly 12,000 final predictions and 31 December predictions. The scorer checks IDs, column structure, finite positive rates and fixed December inputs; it does not calculate predictive accuracy. Large errors on extreme prices remain a limitation. New cities and future-month changes may reduce generalization, and repeated development-window comparisons can introduce selection optimism.

Evidence: supplied model_comparison_sep_oct.csv, model_comparison_jul_aug.csv, executed 02_final_workflow notebook, original assessment data and scorer-generated December chart. Experiments and scoring were run in the candidate notebook; this report summarizes those saved results.